



HABIB UNIVERSITY

DHANANI SCHOOL OF SCIENCE AND ENGINEERING

EE251/CE252	Spring'26	HW 2 SOLUTION	Marks 50 Weight 2.5%	Due: As announced
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Important Notes

Preceding the solutions, you must include the following information:

- Your name
- Student ID
- Course ID
- Homework number
- Date and time submitted

Solution should be elaborative, sketch neat circuits wherever required, show clearly all steps of calculation & mention reference (for example formula, law, rule) to strengthen your understanding to solve the electric circuit problems.

Also, please comply with Academic Integrity Policies of Habib University related to plagiarism while submitting this assignment for grading.

Late Submission Policy

As announced in course syllabus.

CLO-2	Calculate response of DT and CT LTI systems to various inputs and categorize LTI systems	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
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Problem 1 [10 pts]

Determine whether the system described by

$$y[n] = nx[an + 1] + 5,$$

where a is an integer and $|a| > 1$, is

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|---------------------|------------------|
| (i) memoryless; | (ii) invertible; |
| (iii) causal; | (iv) stable; |
| (v) time invariant; | (vi) linear. |

Solution:

- (i) Not memoryless (depends on time $an + 1 \neq n$);
- (ii) Not invertible because $y[0] = 0(x[1]) + 5$ we cannot get from $y[n]$ the value of $x[1]$
- (iii) Not causal ($an + 1 > n$ so depends on input value at future time)
- (iv) Not stable—for example, if $x[n] = 1$ is the input (a constant value 1), then $x[n]$ is bounded but the output is $y[n] = n + 5$ which goes to ∞ as $n \rightarrow \infty$
- (v) Not time invariant: $x[n - n_0] \rightarrow nx[an + 1 - n_0] + 5$ but this $\neq y[n - n_0] = (n - n_0)x[a(n - n_0) + 1] + 5$
- (vi) Not linear: $kx[n] \rightarrow nkx[an + 1] + 5$ but this $\neq ky[n] = k(nx[an + 1] + 5)$

Problem 2 [10 pts]

An LTI discrete-time system has the impulse response

$$h[n] = (0.9)^n u[n].$$

- (a) Determine whether this system is causal.
- (b) Determine whether this system is stable.
- (c) Find the system response to a unit step input $x[n] = u[n]$.

Solution:

- (a) Causal since $h[n] = 0$ for $n < 0$.
- (b) Stable since $\sum_{n=-\infty}^{\infty} |(0.9)^n u[n]| = \sum_{n=0}^{\infty} 0.9^n = \frac{1}{1-0.9} < \infty$.
- (c)

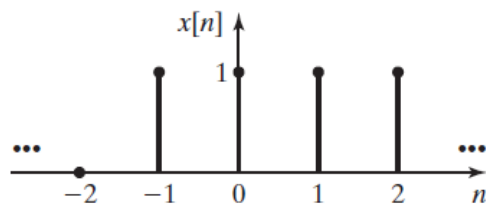
$$\begin{aligned} y[n] &= u[n] * (0.9)^n u[n] = \sum_{k=-\infty}^{\infty} u[k] 0.9^{n-k} u[n-k] \\ &= u[n] \sum_{k=0}^n 0.9^{n-k} = 0.9^n \frac{1-0.9^{-(n+1)}}{1-0.9^{-1}} u[n] \\ &= \frac{0.9^n - \frac{1}{0.9}}{1 - \frac{1}{0.9}} u[n] = \frac{1-0.9^{n+1}}{1-0.9} u[n] \end{aligned}$$

Problem 3 [10 pts]

Consider an LTI system with the input and output related by

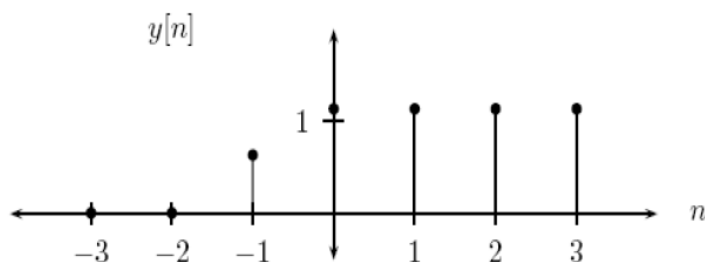
$$y[n] = 0.5x[n - 1] + 0.7x[n].$$

- (a) Find the system impulse response $h[n]$.
- (b) Is this system causal? Why?
- (c) Find the output of the system to the input shown below



Solution:

- (a) $h[n] = 0.5\delta[n - 1] + 0.7\delta[n]$.
- (b) Yes causal—output only depends on past and present (or, simply note that $h[n] = 0, n < 0$).
- (c) $x[n] = u[n + 1]$,
 $y[n] = 0.5(u[n]) + 0.7(u[n + 1])$



Problem 4 [10 pts]

(a) Consider an LTI system with the output given by

$$y[n] = \sum_{k=0}^{\infty} e^{-2k} x[n - k].$$

Find the impulse response of this system by letting $x[n] = \delta[n]$ to obtain $y[n] = h[n]$.

(b) Is this system causal? Why?

(c) Is this system stable? Why?

Solution:

(a) Let $x[n] = \delta[n]$ then

$$y[n] = h[n] = \sum_{k=0}^{\infty} e^{-2k} \delta[n - k] = \sum_{k=0}^{\infty} e^{-2n} \delta[n - k] = e^{-2n}$$

(b) Causal, since $h[n] = 0$ for $n < 0$

(c) Stable, since

$$\sum_{n=-\infty}^{\infty} |h[n]| = \sum_{n=-\infty}^{\infty} e^{-2n} = \frac{1}{1 - e^{-2}} < \infty$$

Problem 5 [10 pts]

Determine whether *the ideal time delay*

$$y(t) = x(t - t_0)$$

is

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|--------------------------------|-------------------------|
| (i) memoryless, | (ii) invertible, |
| (iii) causal, | (iv) stable, |
| (v) time invariant, and | (vi) linear. |

Solution:

i) not memoryless unless $t_0=0$

ii) invertible: $x(t)=y(t+t_0)$

iii) If $t_0 \geq 0$ it is causal; otherwise not.

iv) stable; the output only takes value of the input so if the input is bounded the output will be too.

v) time invariant: let $y_d(t)$ be the output when $x(t - t_1)$ is the input. $x(t - t_1) \rightarrow y_d(t) = x(t - t_1 - t_0)$ and $y(t - t_1) = x(t - t_1 - t_0)$, so $y_d(t) = y(t - t_1)$.

vi) linear: scaling and adding two inputs $ax_1(t) + bx_2(t)$ gives output $ax_1(t - t_0) + bx_2(t - t_0)$, which is the same output we would get by putting $x_1(t)$ and $x_2(t)$ into the system separately and then scaling and adding the outputs.